

September 5, 2026

Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled **“Machine learning in battery management systems: a structured scoping review and classification of 103 peer-reviewed studies, 2019–2026”** for consideration as a review paper in an IEEE Transactions journal.

Machine learning has moved from a promising addition to battery research to a load-bearing component of modern battery management systems (BMS), yet the literature’s self-image rests on validation practices that have never been systematically measured. This manuscript provides that measurement. Based on a Crossref-verified corpus of 103 peer-reviewed journal articles (plus a 14-paper conference sensitivity set from ACC, ECC, CCDC, APEC, VPPC, and ITEC), we classify every study by management function, method family, chemistry, data source, and validation practice, and we quantify how the field has evolved since 2019: from classical feature-based learners through deep sequence models to physics-informed hybrids, optimization-augmented learning, and the first foundation-model studies.

The review offers four contributions that we believe fit the Transactions’ readership: (i) a verified, publicly released corpus and reproducible search pipeline; (ii) a quantified validation-gap analysis (only 9 of 103 studies demonstrate online or fleet-level operation; 23 rest on a single public benchmark); (iii) an evidence-backed “physics-informed validation paradox” (54% vs. 73% strong-protocol share), challenging the assumption that physics priors substitute for out-of-distribution testing; and (iv) two community tools: a safety-and-verification taxonomy for foundation-model components in the BMS stack, and the BMS-ML-RC, a twelve-item CONSORT-style reporting checklist for future machine-learning BMS studies.

We confirm that this manuscript is original, has not been published previously, and is not under consideration by any other journal. All authors have approved the submission and declare no conflicts of interest. Writing-support tools (ChatGPT, Grammarly, QuillBot) were used for language editing only; all technical content, analysis, and interpretations are the authors’ own. The complete classified corpus, the conference sensitivity set, and all analysis scripts are publicly available at <https://github.com/touhidsiddiqueeraj-bit/-structured-scoping-review-and-classification>.

Thank you for your consideration.

Sincerely,

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On behalf of all authors
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